LAB 4: ACTIVITY DIAGRAM

Digital Health Management System (DHMS)

End-to-end workflow: appointment booking, consultation, diagnostic reports and follow-up

Patient
DHMS / System
Receptionist / Clinician
Laboratory Staff
Decision

PATIENT
Open the portal and log in or register

DHMS
Authenticate identity and check role permissions

Credentials valid?
No
DHMS
Show authentication error; allow retry
retry
Yes

PATIENT
Choose provider, service and preferred date

DHMS
Retrieve available slots from provider schedule

Slots available?
No
DHMS
Show alternative dates or available slots
retry
Yes

PATIENT
Select a slot and submit booking request

DHMS
Re-check slot availability atomically

Slot still free?
No
DHMS
Display booking conflict and refresh availability
retry
Yes

DHMS
Create appointment ID, set status and send confirmation

PATIENT
Arrive at the clinic on the scheduled date

RECEPTIONIST
Check in patient and update appointment status

CLINICIAN
Open appointment and load authorized patient context

CLINICIAN
Conduct consultation; record notes, diagnosis and care plan

Lab test required?
Yes
CLINICIAN
Create a laboratory test order
LABORATORY STAFF
Perform test and upload report
DHMS
Validate report metadata and store with provenance
CLINICIAN
Review diagnostic result
No

Continue care

CLINICIAN
Complete care plan and issue prescription if indicated

DHMS
Finalize encounter, record audit event and queue follow-up

PATIENT
Access authorized reports, prescriptions and follow-up details

Main success path with authentication, availability and optional diagnostic-test branches. Other appointment outcomes include rescheduled, cancelled and no-show.

Based on DHMS SRS v1.0, Sections 4.3-4.7 and Appendix B.3.